



Application of XGBoost algorithm in the optimization of pollutant concentration

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ABSTRACT

With the rapid development of the economy, urban pollution has become a hot issue of human concern, and people are put forward higher requirements for the accuracy of pollutant concentration simulations. Based on the the WRF-Chem model simulation results and Beijing environmental monitoring data, this study constructed the XGBoost algorithm through the process of data cleaning, feature selection and super parameter optimization, and the optimized simulation of PM_{2.5} and O₃ concentrations in Beijing was then carried out based on the constructed algorithm. The results showed that the XGBoost algorithm can better capture the spatial and temporal variation patterns of pollutant concentrations, and has a greater improvement on the simulation results of the WRF-Chem model, and that the XGBoost algorithm shows better optimisation results in urban areas compared to suburban areas. In addition, the overall analysis of the features of PM_{2.5} and O₃ concentrations based on the SHAP value theory showed that the time series and periodic features, aerosol ion concentration of Sodium (NAAJ) and Nitrate concentration (NO3AJ) were the important features affecting the prediction of PM_{2.5} concentration by the XGBoost algorithm, and the most important factor affecting O₃ concentration is temperature. PM_{2.5} and O₃ concentrations were divided into three levels, and several samples were selected for single sample analysis. The analysis showed that at low pollutant concentration, most of the features made negative contributions to the concentration prediction, while at high concentration, most of the features made positive contributions. The contribution values of different features varied greatly and were unevenly distributed. The results of prediction were basically composed of a few features with large feature contributions, and the feature contributions of the same feature to different concentration prediction results were also different. Moreover, the XGBoost algorithm was used to optimize the concentration of pollutant at each grid point in Beijing, and a set of pollutant concentration data set with spatial resolution of 6 km and time resolution of 1 h covering the whole Beijing was established, and the optimized spatial distribution of pollutant concentration was closer to the spatial distribution of observed concentration than WRF-Chem simulation. At last, compared with SVR, LR, DTR and RF algorithms, XGBoost algorithm was better than other statistical algorithms in optimising PM_{2.5} and O₃ concentrations. The results of this study provided a new idea for an in-depth analysis of optimization principle of algorithm model for air pollution and a quantitative study of the influencing factors.

1. Introduction

Since the end of the last century, along with the decrease of the concentration of primary pollutants such as NO₂, PM₁₀ and SO₂, there has been a turnaround in the characteristics of air pollution in China. Air

pollution has gradually shifting from primary pollution to secondary pollution dominated by atmospheric fine particulate matter PM_{2.5} and the atmospheric strong oxidant O₃. Although the national average annual concentration of PM_{2.5} in 2018 reduced by 45.8% to 39 µg/m³ compared to 2013, the average annual concentration of PM_{2.5} in typical

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areas of eastern China, such as Beijing-Tianjin-Hebei (BTH), Fenwei plain and Yangtze River Delta, is still high (Gao et al., 2017; Zhai and Chen, 2018; He et al., 2020; Xue et al., 2020). In addition, O₃ pollution has become increasingly prominent in China in recent years, especially in the BTH region, which has become the most important air pollutant and seriously affected human health (Wang et al., 2021; He et al., 2022). Therefore, the simulation of the secondary pollutants PM_{2.5} and O₃ is important for clarifying the causes of air pollution, improving pollution prediction and scientific prevention and control.

At present, there are two main methods to simulate air pollutants: atmospheric numerical model and statistical prediction method (Xiao et al., 2017; Chen et al., 2018; Ke et al., 2022; Liu et al., 2021; Lu et al., 2021; Sicard et al., 2021). Existing studies show that the prediction and simulation of pollutants by numerical models are based on the theory of atmospheric dynamics, using a series of partial differential equations to simulate various physical and chemical processes of air pollutants. Through the use of a large number of approximate theories and equation simplification, leading to some uncertainty in the numerical model itself (Lynch, 2007; Stern et al., 2008). In addition, the model simulation process is easily affected by the initial meteorological field, boundary conditions and various emission source (Zhao et al., 2008; Fu et al., 2013). Therefore, the existing atmospheric numerical models have some uncertainties for the simulation of atmospheric pollutant concentration.

In recent years, with the popularity of artificial intelligence algorithms, more and more researchers began to use statistical prediction algorithms to simulate air pollutants, such as Random Forests (RF), Linear Regression (LR), Decision Tree Regression (DTR), extreme gradient boosting (XGBoost) algorithm, etc. (Lee et al., 2017; Kamińska, 2018; Pan, 2018; Gu et al., 2021; Jiang et al., 2021). Compared with the atmospheric numerical model, the statistical prediction algorithm is simple, efficient and economical. Therefore, the statistical prediction algorithm has attracted more and more attention.

However, the statistical prediction algorithm lacks sufficient explanation of the physical and chemical aspects of the atmosphere, and is difficult to reflect the complex non-linear physical and chemical processes and dynamic mechanisms in the atmosphere (Brokamp et al., 2017; Zhao et al., 2019). At the same time, due to the potential characteristics of statistical prediction algorithm, the relationship found from the data are not necessarily causal, and its black box features in the field of air pollution prediction still exist. Therefore, the achieving interpretability is also the research focus of current algorithms. In addition, the complexity and uncertainty of the data is also one of the factors affecting the simulation ability of artificial intelligence algorithm, resulting in possible physical unreliability and inconsistency (Hou et al., 2018; Wang, 2019). Traditional algorithms only reflect the relative importance of prediction variables according to the prediction ability, but rarely quantify the contribution of algorithm output, so they can not identify and quantify the contribution of pollution sources (Lundberg and Lee, 2017; Gu et al., 2021).

In view of the advantages and disadvantages of the above two methods, this study proposes a method to optimize the simulation of atmospheric pollutants by combining numerical atmospheric models and statistical forecasting algorithms. The WRF-Chem atmospheric chemistry model is selected as the numerical model, and the XGBoost algorithm is selected as the statistical prediction algorithm. Taking PM_{2.5} and O₃ as research objects, the meteorological and pollution data simulated by WRF-Chem model are used as features and input into XGBoost algorithm for pollutant optimization, so as to improve the accuracy of air pollutants in the simulation area. In addition, based on the SHAP value theory, this study calculates the marginal contribution of each feature to the output of XGBoost algorithm, further analyzes the variation pattern of feature contribution with feature value, obtains the relationship between the algorithm prediction and the feature quantity, and then explains the black-box feature of the algorithm from both global and local aspects. Finally, the applicability of XGBoost algorithm in air pollution is explained.

2. Methodology

This study is based on the XGBoost algorithm to optimize the simulation study of pollutant concentrations PM_{2.5} and O₃. The basic process framework is shown in Fig. 1, which is mainly composed of four parts.

(1) The Knowledge Base: The data base of artificial intelligence algorithms. This study mainly combines the data set from two aspects: observation data and WRF-Chem model simulation data. The model simulation data is used as the driver of XGBoost algorithm, and the observation data verifies the accuracy of the algorithm model.

(2) Feature Engineering: In this part, the combined data set will be cleaned according to the time series, include removing error data, filling missing data, etc. and the data will be normalised. Finally, the combined data will be divided into training set, verification set and testing set.

(3) Establish XGBoost algorithm: Select the features of the processed data set according to different pollutants. On the one hand, eliminate the features that have no or even negative contribution to the prediction of pollutant concentration. On the other hand, properly reducing the number of features can save computing resources. Then, the Grid Search with Cross-Validation (GridSearchCV) is used to optimize the hyper-parametric of the algorithm model, so as to improve the optimization effect of the model.

(4) Application: By constructing the optimal XGBoost algorithm, the temporal and spatial optimization of PM_{2.5} and O₃ concentrations is carried out, and the influence of features on pollutant concentration prediction are analyzed.

2.1. WRF-Chem model description

2.1.1. Introduction to WRF-Chem model

The WRF-Chem model is an on-line coupling model of atmospheric dynamics and chemistry developed by the National Oceanic and Atmospheric Administration (NOAA), which achieves the complete coupling of meteorological and chemical in temporal and spatial resolution. The WRF-Chem model includes the feedback mechanism of the impact of pollutant concentration on meteorological conditions, offering the possibility to study the interaction between meteorology and air pollution.

The WRF-Chem model used in this study, in which the gas phase chemical module is developed by Li et al. (2010, 2011a, 2011b, 2012), can be simulated more flexibly by using chemical mechanisms such as CBIV, RAMD2 and SAPRC and the CMAQ aerosol module developed by the U.S. Environmental Protection Agency is also added to the model. In addition, the model simulates organic aerosols by using unconventional VBS (volatility basis set) method (Donahue et al., 2006; Robinson et al., 2007; Li et al., 2011b), and inorganic aerosols by using ISORROPIA version 1.7 (Nenes et al., 1998).

2.1.2. Model setting

The model simulation area is North China, including the entire BTH region and its surroundings. The research area is Beijing, the capital of China, as shown in Fig. 2. The model adopts Lambert projection with the simulation center is located at (38°N, 116°E). The horizontal grid cells is 150 × 150 and the resolution is 6 km. The vertical is divided into 34 layers and the integration time step is 15 s. FNL global reanalysis data is used as the initial field and boundary field for the model to input, which the spatial accuracy is 1° × 1° and the time resolution is 6 h. The model adopts Noah land surface scheme, Lin cloud microphysics scheme, rrtm longwave radiation scheme and Goddard shortwave radiation scheme; The chemical initial/boundary conditions adopts the 6 h output data from MOZART global chemical model. The MEIC list of Tsinghua University is used for anthropogenic emission inventory and the chemical mechanism is SAPRC99. The biogenic emissions are calculated online using the Model of Emissions of Gases and Aerosol from Nature (MEGAN) model. The simulation time is from January 1, 2017 to June 30, 2017 and the specific model settings are shown in Table 1.

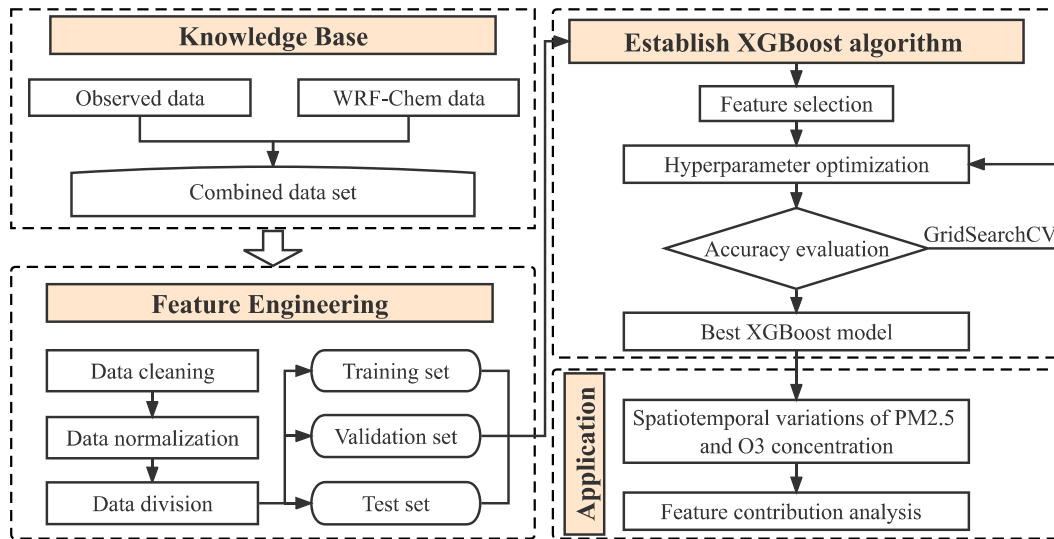


Fig. 1. Basic research framework.

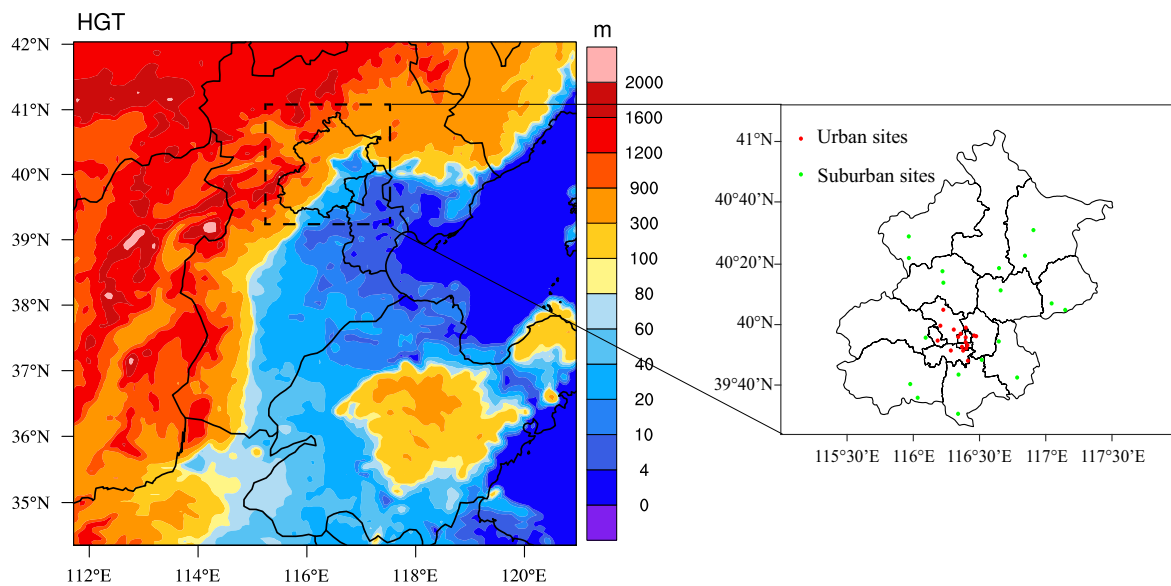


Fig. 2. Model domain with the topography and distribution of observation data sites.

Table 1
WRF-Chem model configuration.

Item	Configuration
Period	2017.01.01–2017.06.30
Region	North China
Domain center	116°E, 38°N
Number of grids	150 × 150
Horizontal resolution	6 × 6 km
Microphysics scheme	Lin
Boundary layer scheme	Eta Mellor-Yamada-Janjic TEK scheme
Surface layer scheme	Monin-Obukhov (Janjic Eta) scheme
Land-surface scheme	Noah
Longwave radiation scheme	rrtm scheme
Shortwave radiation scheme	Goddard scheme
Meteorological boundary and initial condition	NCEP 1° × 1° reanalysis data
Chemical boundary and initial condition	MOZART 6 h output
Anthropogenic emission inventory	SAPRC99 chemical mechanism emissions
Biogenic emission inventory	MEGAN
Spin-up time	2 day

2.1.3. Simulation method

The WRF-Chem model simulation in this study will adopt a time-overlapping step-by-step simulation method. As shown in Fig. 3, the time period to be simulated is divided into several sub-segments for simulation respectively. The first 2 days of each simulation is used as the mode spin-up time, and the next few days are simulated to construct the algorithm input data set. The specific implementation operation is that the simulation start time is 0:00 on the 3rd, 8th, 13th, 18th, 23rd, and 28th of each month and the corresponding end time is 0:00 on the 10th, 15th, 20th, 25th, 30th and the 5th of the following month. This method can eliminate the statistical discontinuity caused by different statistical errors between consecutive runs. Reinitialization by subdividing the long-term continuous integration into shorter ones is an effective method to alleviate the systematic model error problem that may occur in the long-term integration (Du et al., 2014).

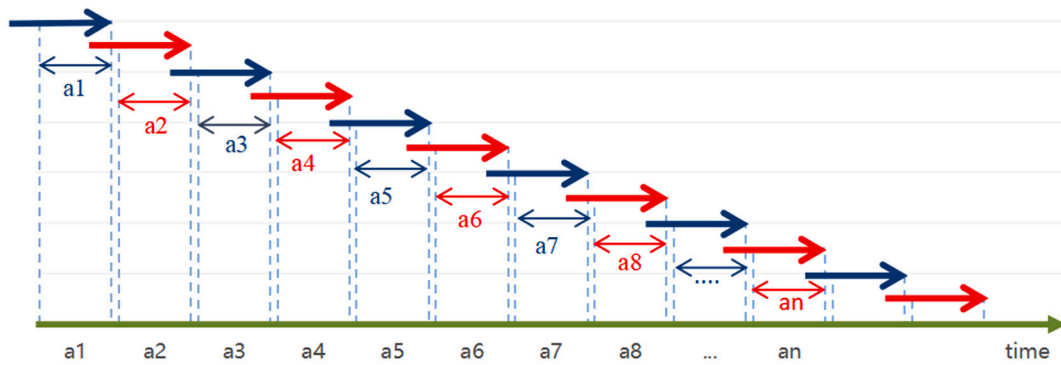


Fig. 3. Time flow chart of WRF-Chem model simulation.

2.2. Data

2.2.1. Observation data

The observation data of this study are from 35 environmental monitoring stations of Beijing Municipal Ecology and Environment Bureau. The distribution of stations is shown in Fig. 1 and the stations name is shown in Table 2, including 17 urban observation stations and 18 suburban observation stations. Due to the error of observation instruments or human factors, some pollutant concentration monitoring values are abnormal values such as missing values or negative values. Therefore, the observation data shall be cleaned according to the time series method. When the number of continuous missing values or abnormal values in the time series is greater than 3, delete the missing or abnormal data of the current site, otherwise use linear interpolation to fill in the data.

2.2.2. WRF-Chem model simulation data

Considering the category of WRF-Chem model simulation data and divided them into meteorological features and pollutant features. In addition, due to the limited amount of model output data, this study also constructs time series features, periodic features and social features as the supplement to the eigenvalues of the input XGBoost algorithm. These three types of features include the time series and periodic variation features of pollutants and the impact of social experience on pollutant concentration, so as to provide more information for the training of the model. We detail each type as follows.

- 1) Meteorological features: The model outputs 28 meteorological features such as T, P and RH, etc.
- 2) Pollutant features: The model outputs 27 pollutant features such as O₃, NO₂, NO, etc.
- 3) Time series features: Time series features is used to describe the short-term fluctuation characteristics of pollutants, i.e. the statistical characteristics (mean and variance) of model simulation pollutants

in their current and n-1 h. In the model simulated pollutant data set $\{x_t\}_{t=1}^T$, x_t represents the model simulated pollutant concentration at time t, then the calculation formula of time series features of the pollutant is:

$$\mu_t^{tem} = \frac{1}{n} \sum_{i=0}^{n-1} x_{t-i}$$

$$\sigma_t^{tem} = \frac{1}{n} \sum_{i=0}^{n-1} (x_{t-i} - \mu_t^{tem})^2$$

where μ_t^{tem} Represents the short-term mean characteristic, σ_t^{tem} Represents short-term variance characteristics. In this study, the values of n are 3, 6, 12 and 24. The time series features of these four different scales of six pollutants (PM_{2.5}, O₃, NO₂, CO, SO₂, PM₁₀) are calculated, and finally 48 short-term time series features of different scales are constructed.

- 4) Periodic features: Since the change of pollutant concentration presents a certain periodicity (such as O₃), the periodic feature is used to describe the long-term periodic fluctuation law of pollutant, i.e. the statistical characteristics at the same time in the past n days:

$$\mu_t^{per} = \frac{1}{n} \sum_{i=0}^{n-1} x_{t-i*24}$$

$$\sigma_t^{per} = \frac{1}{n} \sum_{i=0}^{n-1} (x_{t-i*24} - \mu_t^{per})^2$$

where μ_t^{per} represents the periodic mean characteristic and σ_t^{per} represents the periodic variance characteristic.

Table 2

Stations name of observation data.

Number	Stations name	Number	Stations name	Number	Stations name
1	Chaoyang Aoti	13	Huairouzheng	25	Shunyi Xincheng
2	Chaoyang Nongzhanguan	14	Xicheng Guanyuan	26	Jingnan Quyuadian
3	Dongcheng Dongsi	15	Xicheng Wanshouxigong	27	Jingxibei Quyuadian
4	Dongcheng Tiantan	16	Xizhimen Jiaotongdian	28	Jingxinan Quyuadian
5	Dongsihuan Jiaotongdian	17	Yongdingmen Jiaotongdian	29	Mentougou Longquanzhen
6	Fengtai Huayuan	18	Changpingzhen	30	Miyunzhen
7	Fengtai Yugang	19	Daxing Huangcun	31	Pingguozhen
8	Haidian Beibuxinqu	20	Dingling Duizhaoedian	32	Jingdong Quyuadian
9	Haidian Zhiwuyuan	21	Fangshan Liangxiang	33	Tongzhou Xincheng
10	Haidian Wanliu	22	Shijingshan Gucheng	34	Yanqingzhen
11	Nansanhuan Jiaotongdian	23	Jingdongbei Quyuadian	35	Yizhuang Kaifagu
12	Qianmen Jiaotongdian	24	Jingdongnan quyuadian		

represents the periodic variance characteristic. In this study, the values of n are 3, 7, 14 and 30. The periodic features of the four different scales are calculated for the six pollutants, and finally 48 periodic features of different scales in the long term are constructed.

- 5) Social features: Based on social and seasons to build social features. Considering the influence of working time, working day and season on pollutant, the construction basis of social features is whether it is weekend, whether it is working time and season. Working days and working hours are expressed as yes or no, and seasons are expressed as spring, summer, autumn and winter. After construction, the three category features will be one hot coded (Wang, 2020).

The above five feature types constitute a total of 154 basic features (see Appendix A for the name and description of each features). Because the basic features have different dimensions and units, which will affect the training speed and model accuracy of the model, the Min-Max Normalization method is used to transform these features into the range of [0,1], so as to make the data of different dimensions comparable and improve the speed of model optimization. The normalization formula is as follows:

$$X^* = (X - \text{Min}) / (\text{Max} - \text{Min})$$

where: Min is the minimum value of features; Max is the maximum value of features; X is the data to be processed; X^* is the output result after normalization.

The data set of XGBoost algorithm is composed of the observation data of 35 stations and 154 features output and calculated by WRF-Chem. we divide each data set by a station ratio of 27:4:4 for training, validation and testing.

2.3. Theoretical basis of XGBoost algorithm

XGBoost (eXtreme Gradient Boosting Decision Tree) is an improvement of the Gradient Boosting Decision Tree (GBDT) algorithm proposed by Chen and Guestrin (2016) of the University of Washington. The algorithm aims to improve the running speed and accuracy of the algorithm model, so it is widely used in various industries. The core of the algorithm is as follows:

Assuming that the model has k decision trees, the integrated model can be expressed as follows:

$$\hat{y}_i = \sum_{k=1}^k f_k(x_i), f_k \in F$$

where: F is the set of regression trees, and f is a regression tree in the set.

The main idea of the algorithm is that each update is based on the prediction results of the previous model. By adding a new tree f to fit the residual error between the prediction results of the previous tree and the real value, a new model is formed, and the new model is used as the basis for the next model learning. The details are as follows:

$$\hat{y}_i = 0\hat{y}_i^{(1)} = \hat{y}_i^{(0)} + f_1(x_i)\hat{y}_i^{(2)} = \hat{y}_i^{(1)} + f_2(x_i) \dots \hat{y}_i^{(t)} = \hat{y}_i^{(t-1)} + f_t(x_i)$$

Where: $\hat{y}_i^{(t)}$ represents the prediction result of the t -th time, $\hat{y}_i^{(t-1)}$ represents the prediction result of the previous ($t-1$), and $f_t(x_i)$ represents the residual fitting value by the newly added regression tree.

Obviously, the goal of prediction is to make the predicted value $\hat{y}_i$ of the number group as close as possible to the true value y_i , and the greatest possible generalization capability is required. Therefore, mathematically speaking this is a fonctionnelle optimization problem, so the objective function is simplified as follows:

$$\text{obj}^{(t)} = l(y_i, \hat{y}_i^{(t)}) + \Omega(f_i)$$

where: The first term $l(y_i - \hat{y}_i^{(t)})$ represents the error function, also

called the loss function. The error function to maximize the fit of the sample through continuous learning of the model to obtain the minimum difference between the prediction result and the prediction target. The second term $\Omega(f_i)$ represents the regularization item. The regularization item defines the complexity of the tree and increases the stability of the model by continually simplifying it with the following formula:

$$\Omega(f_i) = \gamma T + \frac{1}{2} \lambda \sum_{j=1}^T w_j^2$$

T represents the number of leaf nodes in the tree and the second part represents the L2 modulus square of the weight function w of the leaf nodes in the tree, i.e. L2 regularization of w , which is equivalent to adding L2 smoothing to the score for each leaf node, with the aim of avoiding overfitting.

2.4. Feature contribution interpretation analysis

The XGBoost algorithm as a machine learning algorithm with black box properties, model evaluation plays the most critical role in explaining the model. In this study, a pollutant concentration optimization model was developed based on the XGBoost algorithm, and the concentrations of 35 stations in Beijing were optimized and analyzed. However, the optimisation process is hidden in the model structure. The regression tree for each new addition to the model has no way of knowing which characteristic variables were selected for pollutant concentration optimization and the contribution of these characteristic quantities to the pollutant optimization criteria. In addition, the regression tree is more complex due to the random sampling of feature data and samples by the XGBoost algorithm and the large feature data and sample size data, thus exacerbating the difficulty of parsing the XGBoost algorithm.

This study is based on the SHAP (SHapley Additive exPlanations) values proposed by Lundberg et al. (Lundberg and Lee, 2017; Lundberg et al., 2018) as an interpretation of the black box properties of the XGBoost algorithm model. SHAP originates from cooperative game theory, for each sample of the algorithmic model produces a predictive value, and the contribution that each feature quantity makes to the predictive value of the sample is called SHAP value, and the formula is expressed as

$$y_i = y_{\text{base}} + f(x_i, 1) + f(x_i, 2) + \dots + f(x_i, j)$$

where: y_i represents the final predicted value for each sample; y_{base} represents the model baseline score, i.e. the mean of the predictions of all samples on the target variable; $f(x_{i,j})$ represents the predicted SHAP value of the j -th characteristic variable of the i -th sample for the target, and the sample prediction is ultimately expressed as the sum of the model baseline score and the contribution score of each characteristic variable. When $f(x_{i,j}) > 0$, it means that the j -th feature of the i -th sample contributes positively to the prediction of the target; When $f(x_{i,j}) < 0$, it means that the j -th feature of the i -th sample has the opposite effect on the prediction of the target, i.e. negative benefit contribution. Therefore, the SHAP value not only gives the magnitude of the sample feature quantity's prediction of the target, but also reflects the positive or negative contribution of each feature quantity to the target prediction. SHAP theory is an ex-post approach to explaining models. The core idea is how to calculate the marginal contribution of each feature to the model output and then explain the black box properties of the algorithmic model both globally and locally. The SHAP method constructs an additive explanatory model that treats all feature quantities as contributors to the target prediction. Based on the SHAP theory, we can calculate the characteristic contribution at different eigenvalues, analyse the variation pattern of the contribution with eigenvalue, and finally obtain the possible relationship between model prediction and eigenvalue to achieve the applicability of the XGBoost algorithm in the interpretation of atmospheric chemistry models.

3. Results and Discussion

3.1. Modeling of the XGBoost algorithm

3.1.1. Feature selection

Figure 4 shows the result of feature selection analysis of $PM_{2.5}$ and O_3 by the XGBoost model algorithm. The number of features (n) and the calculated values of the statistics (the calculation formula of evs , mae , mse , $meae$ and r are shown in Appendix B) are different for the prediction of pollutant concentrations with different thresholds of participation in the algorithm. As can be seen from the figure, the algorithm's score for $PM_{2.5}$ concentration optimization results is relatively stable between 30 and 154 feature numbers; When the number of features is less than 30, R^2 and evs start to decrease gradually, and mae , mse and $meae$ start to increase gradually, i.e. the deviation of the optimization result starts to increase. Considering the saving of computational resources, the optimal number of features for the optimization of $PM_{2.5}$ concentration is therefore chosen as $n = 30$. Similarly, the results are more reasonable when the number of features $n = 50$ in the scoring of the optimization results for O_3 concentration. By the way, Compared with other feature selection methods (Zeng et al., 2020), firstly, our method selects more statistical indicators to measure the impact of features on the algorithm. Secondly, this method takes each feature importance score as a threshold for feature selection, which can better reflect the impact of feature variables on the algorithm.

3.1.2. Hyperparameter optimisation

The parameters of a machine learning model will largely influence the performance of the model. Therefore, in order to construct the optimal XGBoost algorithm model, the parameters of XGBoost need to be optimized. $n_estimators$, $learning_rate$, max_depth , min_child_weight , $subsample$, $colsample_bytree$, $gamma$, $lambda$ and $alpha$ are 9 important parameters in the XGBoost algorithm model that need to be optimized.

Take $PM_{2.5}$ concentration optimization as an example, using grid search method to optimize parameters. First, $n_estimators$ is used as the optimization parameter and all other parameters to be evaluated are set to default values, Fig. 5(a) shows the optimization results. It can be seen that as the spanning tree increases, the model scores higher and higher. When $n_estimators$ is 40, the model score reaches a maximum of 0.804, after which the model score starts to decrease slowly as the tree increases, so set $n_estimators$ to 40. After determining the value of $n_estimators$, continue to optimize the remaining parameters using the grid search method. Set $n_estimators$ to 40, all other parameters to default values, max_depth and min_child_weight together for optimization ,

After several trials the max_depth range was reduced to 8–20 and the min_child_weight range was reduced to 2–10, Fig. 5(b) shows the results of this parameter optimization. When $max_depth = 14$, $min_child_weight = 6$, the highest model score is 0.805; Next, set $n_estimators$ to 40, max_depth and min_child_weight to 14 and 6 respectively, and perform gamma parameter optimization. As shown in Fig. 5(c), it can be seen that the highest model score is 0.80544 when $gamma = 0.4$. Follow this method for parameterised optimization of $subsample$, $colsample_bytree$, $lambda$, $alpha$ and $learning_rate$, as shown in Fig. 5(d-f), the highest model score is 0.8077 when $subsample = 0.6$, $colsample_bytree = 0.6$, $lambda = 3$, $alpha = 0.1$, $learning_rate = 0.1$, at this point, the XGBoost algorithm model parameter optimization is completed. Table 3 shows the parameter setting scheme of the XGBoost algorithm model for optimising the $PM_{2.5}$ and O_3 concentrations respectively after parameter optimisation.

3.2. Optimized simulation of pollutants $PM_{2.5}$ and O_3 by XGBoost algorithm

Modeling the XGBoost algorithm with more pollutant-optimized characteristics after feature engineering and parameter optimization. Using this algorithm, two urban sites (Haidian Wanliu and Chaoyang Nongzhanguan, abbreviated as HW and CN) and two suburban sites (Huairouzheng and Yizhuang Kaifu, abbreviated as HZ and YK) were randomly selected as test set sites for the optimal simulation of pollutant concentrations.

3.2.1. $PM_{2.5}$ optimization simulation research

Fig. 6 shows the distribution of observed $PM_{2.5}$ concentrations, WRF-Chem simulated values and XGBoost optimized values over a time series at the four test sites. As can be seen from the figure, both the WRF-Chem model and the XGBoost algorithm can simulate the overall trend of $PM_{2.5}$ concentrations quite well. The main manifestations are heavier pollutants in winter and lighter pollution in spring and summer, with a more obvious trend of pollutant concentrations over time. However, the XGBoost algorithm optimization results are better than the WRF-Chem model simulation. The XGBoost algorithm can better capture the spatio-temporal series variation of $PM_{2.5}$ concentration, especially in the period of mid-late April 2017, for the abnormal surge in $PM_{2.5}$ concentration, which was not simulated by the WRF-Chem model, but the XGBoost algorithm was able to capture this trend well, showing the better generalizability of the algorithm. Table 4 shows the observed, modelled and optimized statistics for $PM_{2.5}$ concentrations at the four stations calculated. It is evident from the data that the pollutant concentrations calibrated by the XGBoost algorithm have a higher

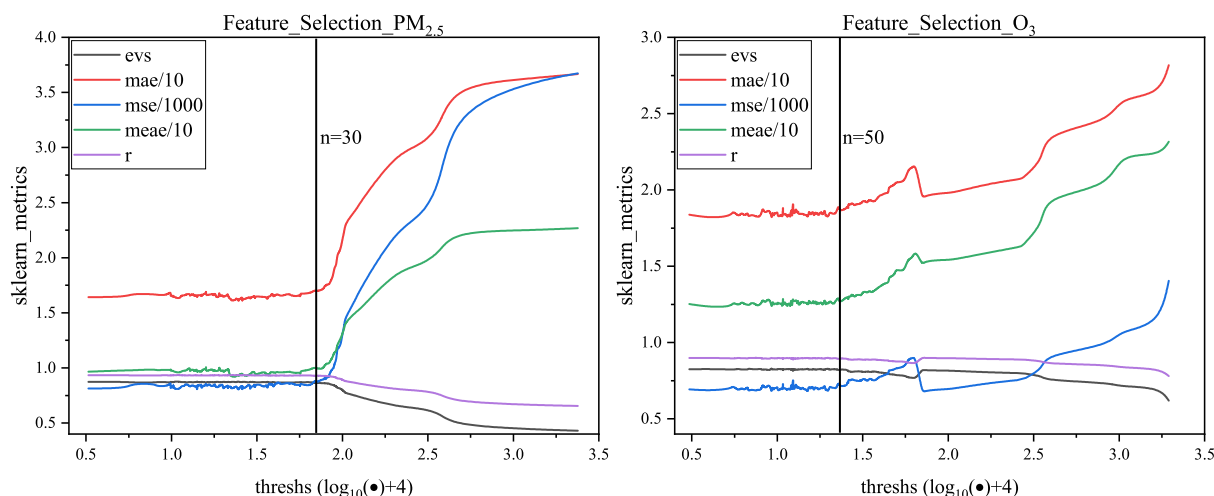


Fig. 4. Feature selection evaluation index for $PM_{2.5}$ and O_3 .

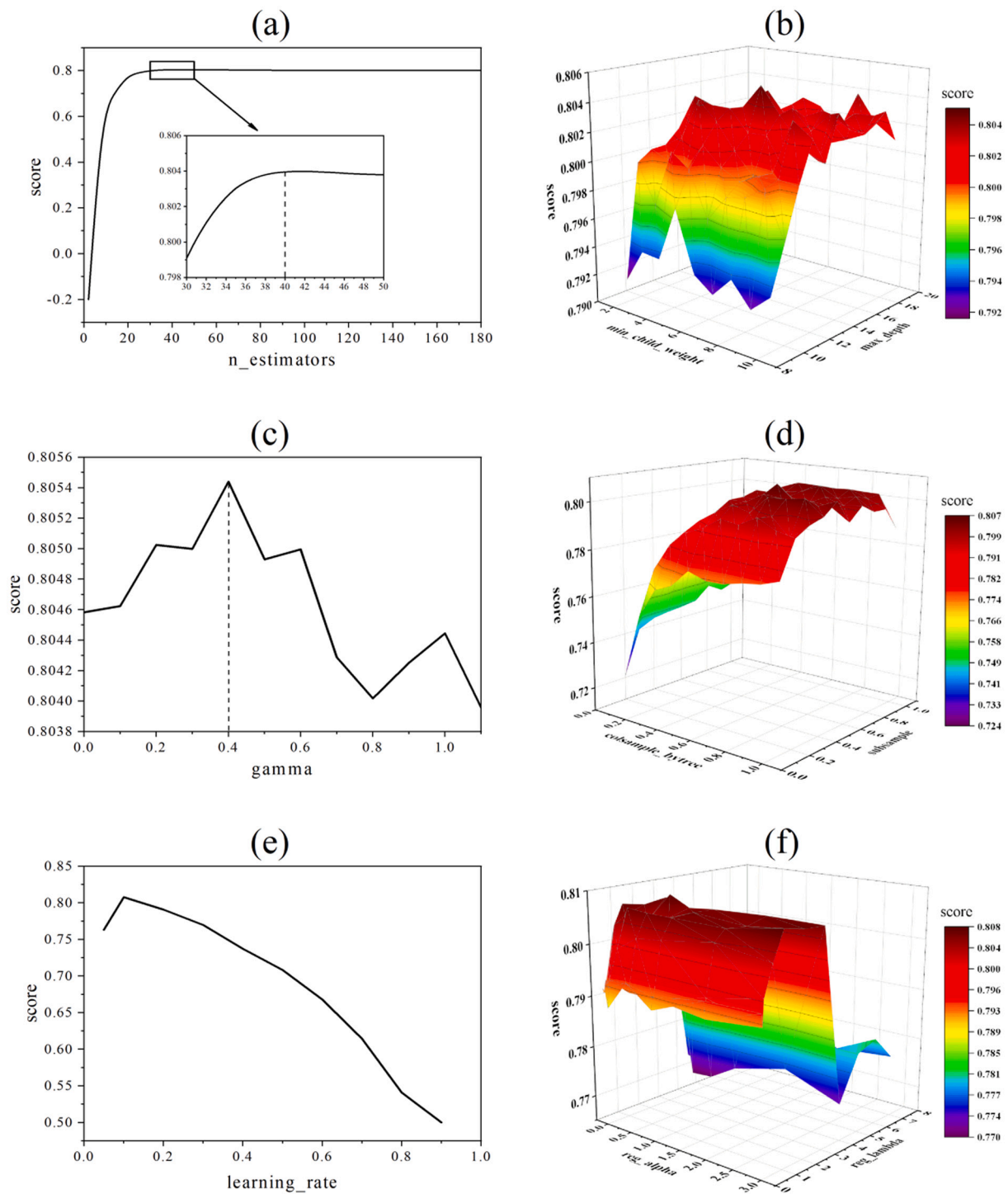


Fig. 5. Hyperparameter optimization results of XGBoost algorithm.

correlation and smaller errors with the observed data, and the IA is also all above 0.9, thus showing a stronger pollutant optimisation capability. Compared with most previous $PM_{2.5}$ simulations based on satellite or other algorithms (Gui et al., 2020; Zeng et al., 2021), although the feature variables and algorithms are different, we have a good simulation effect on the concentration of $PM_{2.5}$. In addition, comparing the optimization results for urban and suburban sites shows that urban sites are better optimized than suburban sites, with correlation coefficients above 0.95, while showing smaller errors. This may be due to the small size of urban areas and the dense distribution of observation data sites, while suburban sites are more sparsely distributed and less correlated

between sites, resulting in poorer optimisation.

3.2.2. O_3 optimization simulation research

Fig. 7 shows the time series distribution of the O_3 concentration observations, WRF-Chem simulations and XGBoost optimized values at the four test sites. As can be seen from the figure, there is a clear seasonal variation in O_3 concentrations, with low concentrations in winter and spring and significantly higher concentrations in summer. In terms of time series, the XGBoost algorithm model shows better optimization overall and better matches the observations. However, for the CN site, both the WRF-Chem model and the XGBoost algorithm underestimated

Table 3Optimal selection of parameters for PM_{2.5} and O₃.

Parameter	Optimal selection (PM _{2.5})	Optimal selection (O ₃)
n_estimators	40	100
max_depth	14	14
min_child_weight	6	2
gamma	0.4	0.2
subsample	0.6	0.6
colsample_bytree	0.6	0.9
alpha	3	0.7
lambda	0.1	1
learning_rate	0.1	0.009
score	0.8077	0.8383

peak O₃ concentrations to varying degrees during the summer period, while the other sites were more accurate at the peak. Table 5 shows the observed, modelled and optimized statistics for PM_{2.5} concentrations at the four stations calculated. It can be seen that the WRF-Chem model is more accurate for O₃ than PM_{2.5}, with correlation coefficients above 0.7. Meanwhile, the correlation coefficients of the XGBoost algorithm are above 0.9 after optimization of O₃ concentration; over 26% improvement in correlation coefficient after optimisation; over 28% reduction in root mean square error, showing good optimisation results. Similar to PM_{2.5} optimization, XGBoost shows better optimization for urban areas than suburban areas.

3.3. Feature interpretation and analysis of the XGBoost algorithm model

3.3.1. The feature interpretation analysis for PM_{2.5} concentration optimisation

Fig. 8 shows the impact of each feature of the XGBoost algorithm on the optimization of PM_{2.5} concentration, and the ordinate from top to bottom indicates the ranking of global feature importance from high to low. It can be seen that the most important feature factor influencing the optimization of PM_{2.5} is the short-term mean feature of the model simulated PM_{2.5} concentration for the first 6 h and 12 h. When the eigenvalues $u_{tem_PM25_6}$ and $u_{tem_PM25_12}$ are higher, the SHAP value is higher, indicating that the high value of the feature contributes positively to the optimization of PM_{2.5}. While the eigenvalue $u_{tem_PM25_6}$ and $u_{tem_PM25_12}$ are lower, the feature inhibits PM_{2.5} generation and the positive benefit contribution of the feature is

greater than the negative benefit contribution. In addition, the temporal and cyclical features account for the majority of the features that have a significant impact on the optimization of PM_{2.5} concentrations, and therefore the temporal and cyclical features play an important role in the optimization of PM_{2.5} concentrations. In addition to the temporal and periodic characteristics, NAAJ (aerosol sodium concentration) and NO3AJ (nitrate concentration) are the pollutant features that have the most important impact on PM_{2.5} optimization. For NAAJ, it has a positive benefit contribution to PM_{2.5} optimization at high values, and as its concentration value gradually decreases, its feature contribution to PM_{2.5} optimization first gradually increases and then begins to gradually decrease, further becoming a negative benefit contribution, while NO3AJ has a positive benefit contribution to PM_{2.5} at high concentration values and a negative benefit contribution at low values. Fig. 9 shows the interaction effect of the two most important feature factors $u_{tem_PM25_6}$ and $u_{tem_PM25_12}$ on PM_{2.5} concentration optimisation. As can be seen from the graph, as the eigenvalues increase, the feature contributions also gradually increase. For the first 6 hourly mean PM_{2.5} concentrations, the feature contribution first increases sharply as its value increases. Later, when the normalised value of $u_{tem_PM25_6}$ exceeds 0.16, its feature contribution starts to stabilise gradually, with a break in the distribution of the feature contribution around 0.16, i.e. there is a sudden increase in the feature contribution here. In contrast, for the first 12-h mean PM_{2.5} concentrations, the contribution of the low value feature is small and the contribution of the high value feature is large.

Table 4Statistics calculation of 4 sites before and after PM_{2.5} optimization.

Sites	PM _{2.5}	R	RMSE	MAE	MB	NMB	IA
HW	obs-sim	0.58	72.12	41.30	3.52	0.05	0.75
	obs-xgb	0.96	21.53	13.36	2.72	0.04	0.98
CN	obs-sim	0.60	72.39	40.45	-0.11	-0.00	0.76
	obs-xgb	0.97	21.08	12.42	2.51	0.04	0.98
HR	obs-sim	0.57	64.52	37.21	6.31	0.11	0.74
	obs-xgb	0.89	34.56	20.71	13.14	0.23	0.93
YK	obs-sim	0.68	66.82	40.55	-8.08	-0.10	0.82
	obs-xgb	0.91	36.16	20.58	-5.98	-0.08	0.95

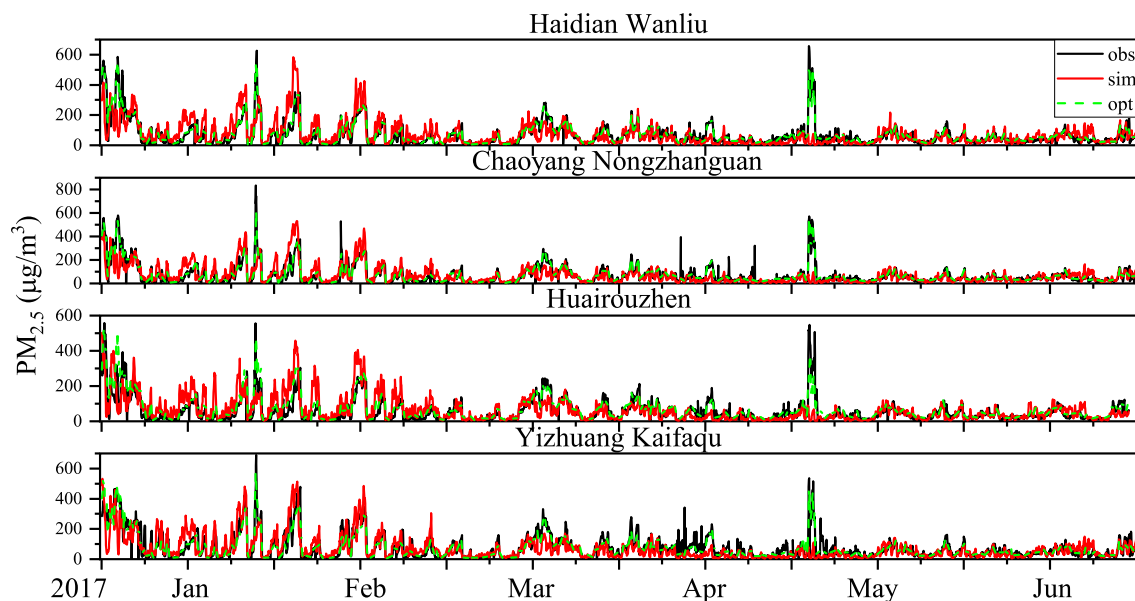


Fig. 6. Results of PM_{2.5} concentration observation (black), simulation (red) and optimization (green) at 4 sites in urban and suburban areas. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

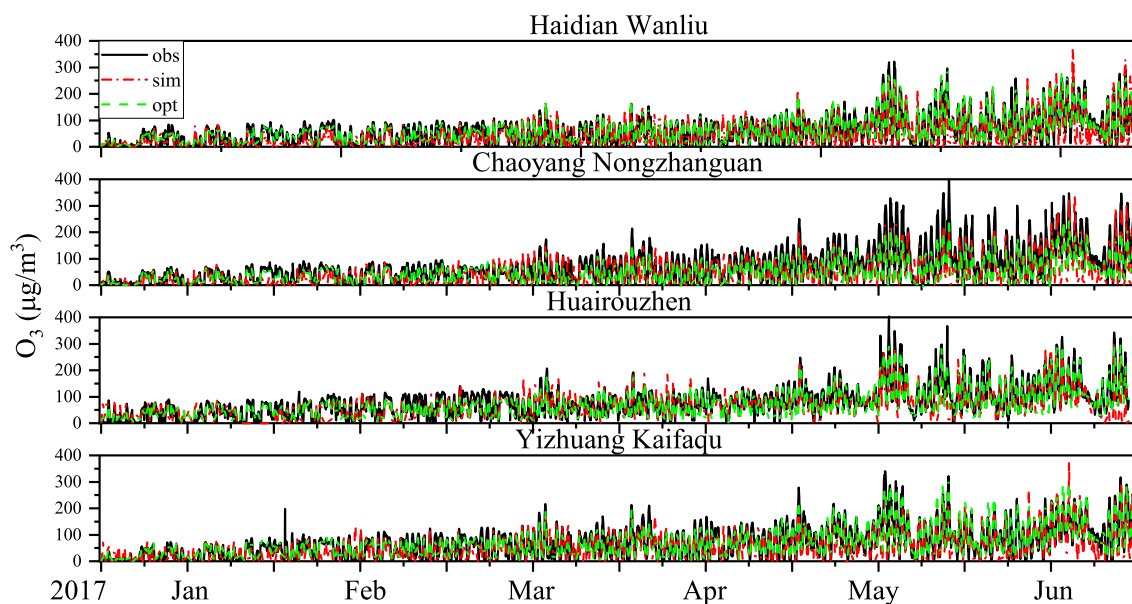


Fig. 7. Results of O₃ concentration observation (black), simulation (red) and optimization (green) at 4 sites in urban and suburban areas. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 5

Statistics calculation of 4 sites before and after O₃ optimization.

Sites	O ₃	R	RMSE	MAE	MB	NMB	IA
HW	obs-sim	0.75	40.01	27.85	−15.43	−0.27	0.85
	obs-xgb	0.96	15.87	11.52	2.45	0.04	0.97
CN	obs-sim	0.75	50.88	35.27	−27.14	−0.39	0.81
	obs-xgb	0.95	36.70	25.62	−23.34	−0.33	0.89
HR	obs-sim	0.71	44.33	31.77	−16.55	−2.1	0.81
	obs-xgb	0.91	26.08	20.01	−10.48	−0.13	0.94
YK	obs-sim	0.78	41.41	29.91	−17.00	−0.24	0.86
	obs-xgb	0.94	20.90	15.55	−1.16	−0.02	0.97

3.3.2. The feature interpretation analysis for O₃ concentration optimisation

Fig. 10 shows the effect of each feature of the XGBoost algorithm on the optimization of O₃ concentration. It is clear from the figure that the most significant factor influencing O₃ concentration is the temperature T, and the higher the temperature the higher the SHAP value, indicating that high temperatures have a greater contribution to O₃ optimization; when the temperature decreases, the SHAP value of temperature for O₃ optimization also decreases, i.e. low temperature has a negative contribution to O₃ optimization. The main causes of this phenomenon are the increase in surface temperature when solar radiation is strong and the gaining effect of strong solar radiation on O₃ production; while when solar radiation is weaker, the temperature decreases accordingly, leading to a decrease in O₃ production. Therefore, temperature can directly affect the production of O₃. Other studies have shown that temperature is the most important meteorological factor as an influence on near-surface O₃ production, which is more consistent with the present study (Otero et al., 2021; Jodzis and Baran, 2022). In addition, the short-term mean features of the model simulated O₃ concentrations for the first 3 h and 6 h are also the main factors influencing the optimization of O₃ concentrations, and basically show a positive contribution, while the positive and negative contributions of the boundary layer height PBLH to the optimization of O₃ concentrations are more balanced. It is worth noting that Working_hour is used as a category feature variable: When working_hour is positive, it means non-working time; When working_hour is negative, it means working hours. The figure clearly shows that the algorithm contributes positively to O₃ predictions during working hours and negatively during non-working hours, indicating that O₃ production is more severely influenced by

day and night, and indirectly reflecting the daily variability of O₃ production. Other feature elements also have varying degrees of influence on O₃ optimisation. Fig. 11 shows the interaction effect of the two most important factors affecting the optimization of O₃ concentration, T and u_tem_O3_3, on the optimization of O₃. When T is low, the SHAP value is negative, reducing the algorithm's prediction of O₃ concentration values; as T exceeds 291.71 K (normalised value of 0.6), the SHAP value of temperature on O₃ production gradually changes from negative to positive with increasing temperature and continues to increase, gradually becoming the most significant factor affecting the predicted value of O₃. The u_tem_O3_3 has similar features, with a large impact on O₃ optimisation and a positive effect contribution at larger values, and a mainly negative effect contribution at smaller values.

3.4. Analysis of the feature contribution of feature data to PM_{2.5} and O₃ concentrations under different classification standards

In order to investigate the feature contribution of the feature data to the output of the XGBoost model at different PM_{2.5} and O₃ concentrations, the daily average PM_{2.5} concentrations and the average hourly O₃ concentrations were classified according to the *Ambient Air Quality Standards*, in the manner shown in Table 6:

Since the model output concentrations are hourly concentrations, the first 24-h average is taken for the pollutant PM_{2.5}. For the 24-h average PM_{2.5} concentration at time t , Y_t can be expressed as:

$$Y_t = \frac{1}{24} \sum_{i=0}^{23} y_{t-i}$$

where: Y_t represents the 24-h average concentration of the pollutant PM_{2.5} at the current moment in time, y_t represents the pollutant concentration at the current moment in the model output; the model output O₃ concentration is the 1-h average concentration.

3.4.1. Feature contribution of feature data to samples with different levels of PM_{2.5} concentration

Two samples were selected within the PM_{2.5} primary concentration standard, with forecast pollutant concentrations of 9.82 µg/m³ and 33.39 µg/m³ respectively, both of which are mildly polluted PM_{2.5} concentrations. Two samples were also selected within the PM_{2.5} secondary concentration standard, with forecast pollutant concentrations

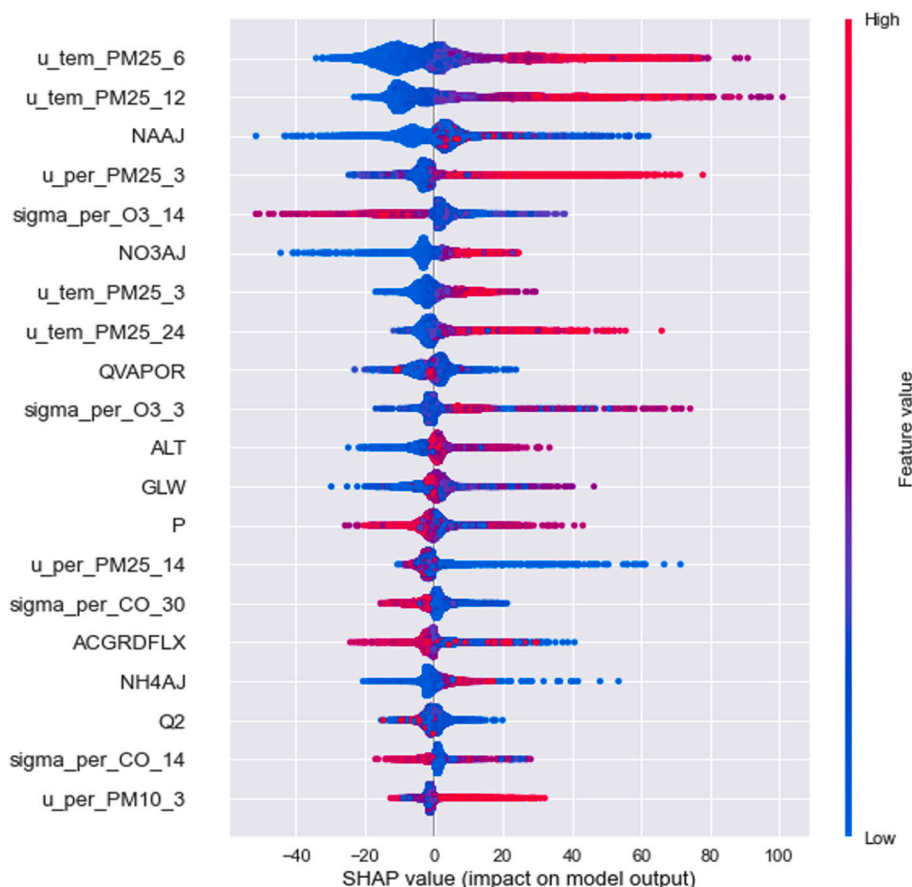


Fig. 8. The feature contribution for PM_{2.5} concentration optimization.

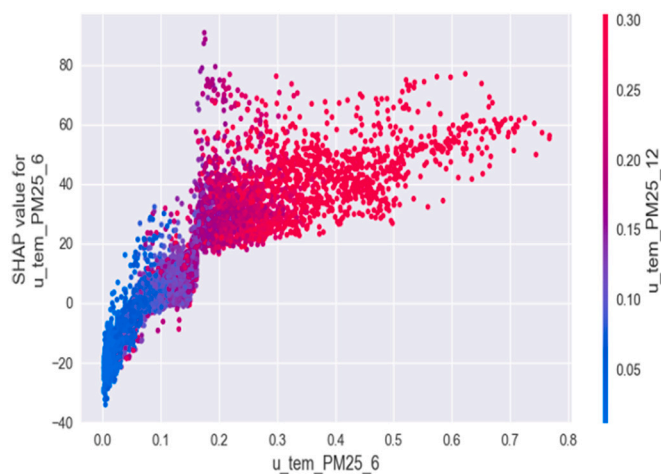


Fig. 9. The influence of the synergy of $u_{\text{tem_PM25_6}}$ and $u_{\text{tem_PM25_12}}$ on feature contribution.

of $47.45 \mu\text{g}/\text{m}^3$ and $62.96 \mu\text{g}/\text{m}^3$. Because of the large range of pollutant concentrations for PM_{2.5} under the tertiary standard, three samples were selected from this range with forecast pollutant concentrations of $90.75 \mu\text{g}/\text{m}^3$, $245.38 \mu\text{g}/\text{m}^3$ and $406.54 \mu\text{g}/\text{m}^3$ respectively. Fig. 12 shows the distribution of feature contributions for model optimization under different PM_{2.5} concentration classification criteria. Each colour block represents the feature contribution of different feature elements to the concentration of a particular sample, and the subscript values representing the eigenvalue of each feature element after normalization.

Through the analysis of the contribution of each feature element in the sample, for a single sample, the contribution values of different features vary widely and are unevenly distributed. Concentration prediction results basically consist of a few features with large feature contributions, and there are differences in the feature contribution values of the same feature for different concentration prediction results. Comparing the feature contributions of each feature element to the predicted pollutant concentrations between these seven samples, it can be seen that the algorithm predicts a mean PM_{2.5} concentration of $69.79 \mu\text{g}/\text{m}^3$, and each feature element has a positive and negative benefit contribution in different sample cases. Positive benefit contributions increase PM_{2.5} concentration projections and negative benefit contributions decrease PM_{2.5} concentration projections. The contribution values of all feature elements to the predicted PM_{2.5} concentrations and the model prediction baseline values are summed to the predicted PM_{2.5} concentration $f(x)$ for the final sample. When PM_{2.5} concentrations are at the primary standard concentration, QVAPOR (water vapor mixing ratio) and GLW (downward long wave flux at ground surface) and the first 12 h mean PM_{2.5} concentration make the main negative contribution to the pollutant concentration, and the main contributor to the positive benefit at NAAJ and the first 14 days variance of O₃ concentrations. With PM_{2.5} concentrations in the secondary standard range, the first 14 days variance of O₃ concentrations and the temporal and periodic characteristics of PM_{2.5} concentrations contribute negatively to the model forecast pollutant concentrations, while the feature elements that contribute positively are mainly NAAJ and NO₃AJ. When PM_{2.5} concentrations are at tertiary standard concentrations, a heavy weather type, the temporal and periodic characteristic values of PM_{2.5} become larger and gradually shift towards a positive benefit contribution, while the NAAJ gradually produces a negative benefit contribution.

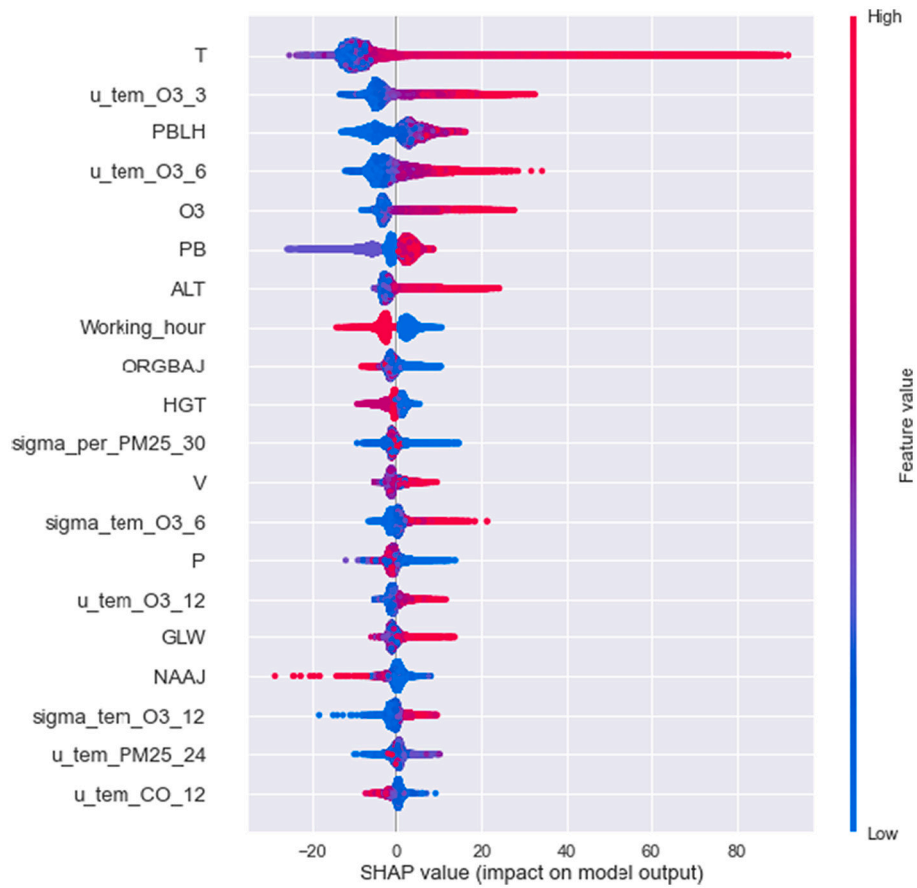


Fig. 10. The feature contribution for O₃ concentration optimization.

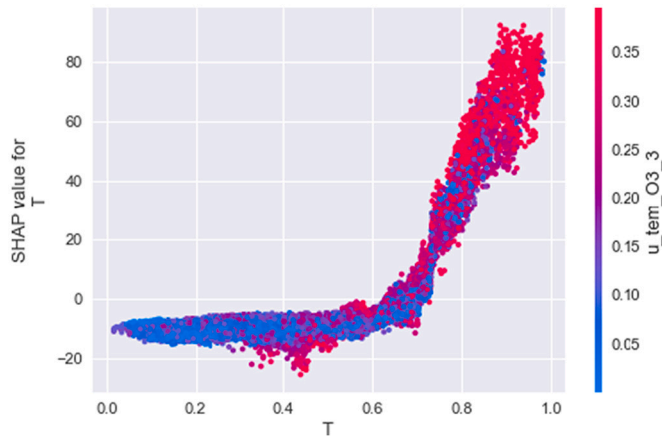


Fig. 11. The influence of the synergy of T and u_{tem}O₃_3 on feature contribution.

Table 6

The concentration standard of PM_{2.5} and O₃.

Level	PM _{2.5} concentration (daily)	O ₃ concentration (hourly)
First	<35 µg/m ³	<160 µg/m ³
Second	<75 µg/m ³	<200 µg/m ³
Third	>75 µg/m ³	>200 µg/m ³

3.4.2. Feature contribution of feature data to samples with different levels of O₃ concentration

Two samples were selected within the O₃ tertiary concentration standard, with forecast pollutant concentrations of 295.19 µg/m³ and 332.21 µg/m³ respectively, both of which are heavy O₃ concentrations. Two samples were also selected within the O₃ secondary concentration standard, with forecast pollutant concentrations of 181.51 µg/m³ and 195.49 µg/m³. Because of the large range of O₃ primary standards, three samples were selected from this range with forecast pollutant concentrations of 6.41 µg/m³, 62.74 µg/m³ and 135.59 µg/m³ respectively. Fig. 13 shows the distribution of feature contributions for model optimization with different O₃ concentration classification criteria. From these seven different concentration prediction samples, it can be seen that the mean value of the model predicted O₃ concentration is 65.07 µg/m³, and each feature element has a positive and negative benefit contribution in different sample cases. The contribution values of all the feature elements to the O₃ concentration prediction and the baseline model prediction sum up to the predicted O₃ concentration $f(x)$ for the final sample. Throughout the seven samples at the same time, the temperature T and the first 3 h time series variation characteristics of O₃ contributed significantly to the prediction of O₃ concentrations. When the predicted concentration is high, the T value is larger and the feature contribution value is also the highest, with SHAP values up to 90, making a positive benefit contribution to the concentration prediction. When the predicted concentration is low, the T value is small and the feature contribution is also dominant, but at this point it makes a negative benefit contribution to the O₃ concentration prediction. The smaller the T value, the greater the negative contribution. Combined with the previous study of the relationship between T and the algorithm's forecast O₃ concentration, T produces a negative benefit contribution when $T < 0.6$ and a positive benefit contribution when $T >$

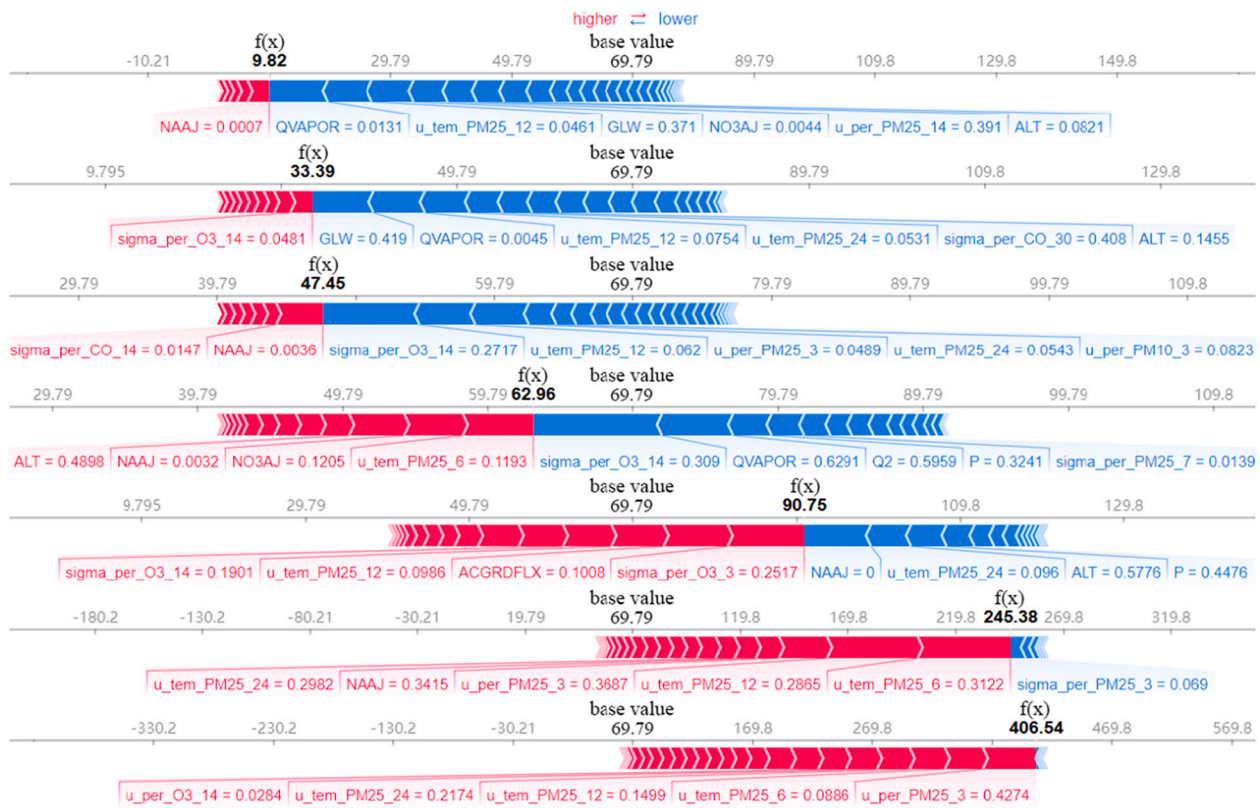


Fig. 12. The distribution of feature contribution of model optimization under different PM_{2.5} concentration standards.

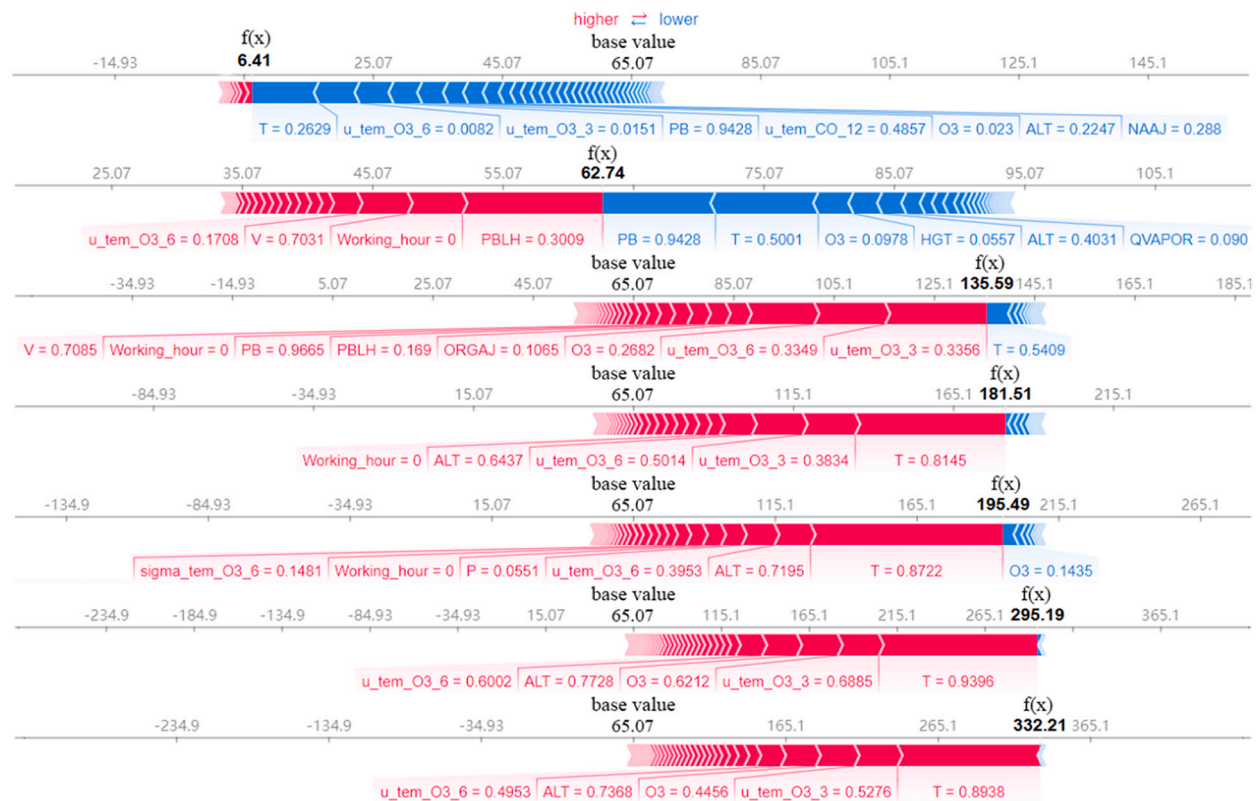


Fig. 13. The distribution of feature contribution of model optimization under different O₃ concentration standards.

0.6, the analysis of a single sample can also lead to this conclusion, forming an echo. In addition, for concentrations predicted to be below $65.07 \mu\text{g}/\text{m}^3$, the negative contribution of feature elements accounts for the major part. For lower predicted concentrations, around $65 \mu\text{g}/\text{m}^3$, the distribution of positive and negative contributions from the feature elements is more balanced. For concentrations predicted above $65 \mu\text{g}/\text{m}^3$, positive contributions from characteristic elements account for the major part. For higher predicted concentrations, and when predicted concentrations exceed $200 \mu\text{g}/\text{m}^3$ to reach heavy pollution, the feature elements are essentially all positive benefit contributions.

3.5. 6 km pollutant concentration data set for the Beijing area

In order to obtain the optimized spatial distribution of pollutant concentrations in the Beijing region, this study uses the established XGBoost algorithm, using 35 station observation data as training data and 31×31 grid points with a simulated resolution of 6 km in the WRF-Chem model in the Beijing region as the test set, with every 4 grid points as a group for cyclic optimization, a total of 241 sets of optimization are needed to obtain the optimized data set corresponding to the grid point data of the WRF-Chem model simulating the Beijing region.

Figs. 14 and 15 show the spatial distribution of $\text{PM}_{2.5}$ and O_3 concentrations in Beijing during high pollution moments after observation, WRF-Chem simulation and XGBoost algorithm optimization, respectively. Where figure (a) shows the spatial distribution of observed pollutant concentrations at the 35 stations interpolated by the Radial Basis Function (RBF) method; The figure (b) shows the spatial distribution of pollutant concentrations at a grid resolution of 6 km after optimization by the XGBoost algorithm; Figure (c) represents the spatial distribution of pollutant concentrations simulated by the WRF-Chem model. As can be seen from the figure, the spatial distribution of pollutant concentrations after optimization by the XGBoost algorithm is closer to the observed results. The WRF-Chem model was less effective in simulating pollutant concentrations during the study period: it shows that a low simulation of $\text{PM}_{2.5}$ concentrations in the western and northern parts of Beijing, a high simulation in the southwest, and a low simulation of O_3 concentrations overall, especially in the eastern part of Beijing. The spatial distribution of pollutants shows that the XGBoost algorithm is able to calibrate the WRF-Chem model simulation results well, bringing the pollutant concentrations closer to the observed concentrations. Due to the concentrated distribution of observation stations and the small number of stations around the Beijing area, the pollutant concentrations optimized by the XGBoost algorithm also show a certain degree of error. For example, the optimization of $\text{PM}_{2.5}$ and O_3 concentrations in the southwest and northeast of Beijing, respectively, is less effective, with no significant change in $\text{PM}_{2.5}$ concentrations and low O_3 concentrations after optimization.

3.6. Effectiveness of different algorithmic models for pollutant concentration optimization

In addition, other commonly used algorithmic models were computed to optimize the simulation of $\text{PM}_{2.5}$ and O_3 pollutant concentrations, including SVR (Support Vector Regression), SGDR (Stochastic Gradient Descent Regression), LR (Linear Regression), DTR (Decision Tree Regression) and RF (Random Forest). The calculation of the final statistics shows that the XGBoost algorithm shows the best optimization, with the highest correlation coefficient and the smallest error, which is more consistent with the results of some scholars' studies (Pan, 2018; Ma et al., 2020). Secondly, the RF algorithm also performs better, and it is also worth mentioning that the SVR algorithm optimises O_3 better than $\text{PM}_{2.5}$ (See Table 7).

4. Conclusion

This study adopts the XGBoost algorithm combined with the atmospheric chemistry model WRF-Chem, using simulated data of atmospheric pollutants and station monitoring data in the Beijing area, to construct an XGBoost algorithm model which more suitable for pollutant concentration optimization. At the same time, the feature importance of the algorithm model for the optimization of atmospheric pollutant concentrations is discussed and analyzed based on SHAP value theory, both in terms of the total and individual samples. The results of the study is summarised below:

- (1) According to the pollutant concentration monitoring data from 35 stations in Beijing, combined simulation of meteorological conditions and pollutant concentrations in and around Beijing with the WRF-Chem atmospheric chemistry model, build a model XGBoost suitable for $\text{PM}_{2.5}$ and O_3 concentration optimization through the process of data cleaning, feature selection and hyperparameter optimization, and apply it to $\text{PM}_{2.5}$ and O_3 concentration optimization. The algorithm model can well capture the spatial and temporal variation patterns of $\text{PM}_{2.5}$ and O_3 concentrations, and the correlation coefficients between the optimized pollutant concentrations and observations are basically above 0.9, and the algorithm has a better optimization effect for urban areas, with correlation coefficients above 0.95.
- (2) Based on SHAP value theory, the overall analysis of the features of the XGBoost algorithm for $\text{PM}_{2.5}$ and O_3 concentration optimization shows that: Temporal and periodic characteristics, aerosol sodium ion concentration and nitrate concentration are important feature factors affecting the prediction of $\text{PM}_{2.5}$ concentrations by the XGBoost algorithm, while the most significant feature factor influencing the optimization of O_3 concentrations

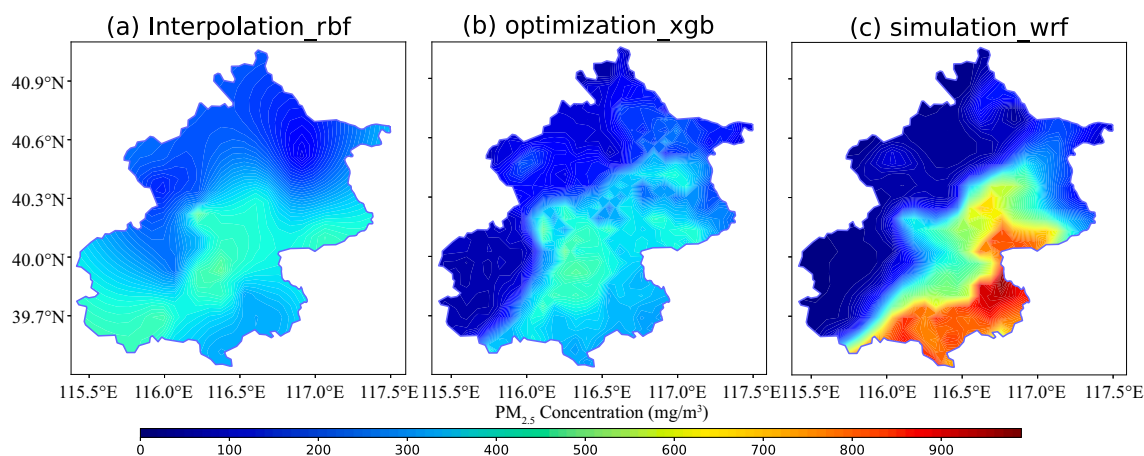


Fig. 14. $\text{PM}_{2.5}$ observed, optimized and simulated concentration spatial distribution map in Beijing.

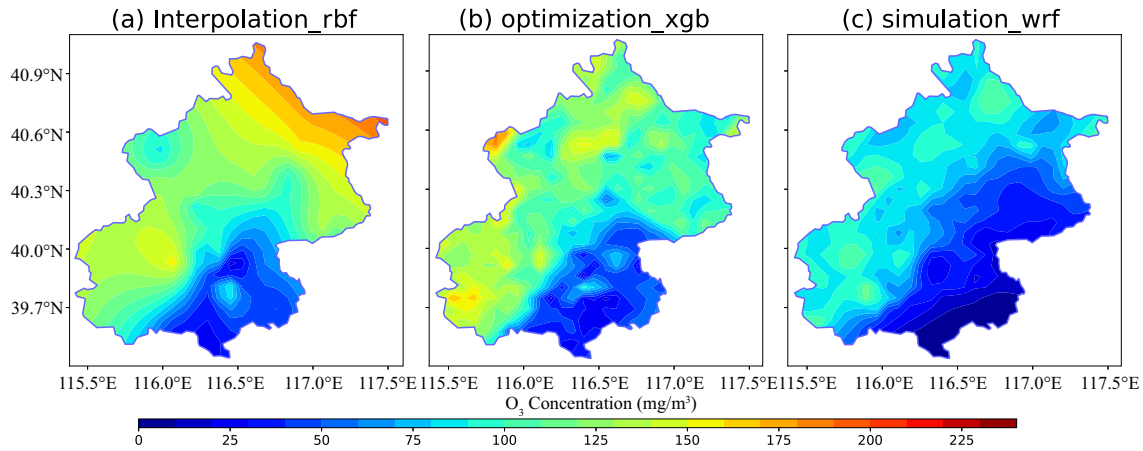


Fig. 15. O₃ observed, optimized and simulated concentration spatial distribution map in Beijing.

Table 7
Statistics for the comparison between XGBoost and other algorithms.

Model	PM _{2.5}			O ₃		
	R	RMSE	MAE	R	RMSE	MAE
WRF-Chem	0.57	72.12	41.30	0.74	40.02	27.85
SVR	0.73	55.40	25.75	0.91	22.99	17.01
SGDR	0.76	52.25	31.76	0.88	25.19	19.27
LR	0.78	50.27	31.16	0.89	24.65	18.95
DTR	0.90	36.47	17.49	0.89	25.81	17.31
RF	0.96	21.52	13.31	0.96	16.02	11.57
XGBoost	0.97	20.80	11.49	0.96	15.82	11.17

is temperature, and its importance is much greater than the contribution of other feature elements. In addition, by dividing PM_{2.5} and O₃ concentrations into three levels, the feature contribution of individual samples to the algorithm model at different pollution concentrations is investigated, and the results show that the baseline values of XGBoost algorithm for PM_{2.5} and O₃ concentration prediction are 69.79 µg/m³ and 65.07 µg/m³ respectively. The temporal and periodic SHAP values become larger at increasing PM_{2.5} concentrations and gradually produce a positive benefit contribution, while the NAAJ gradually produces a negative benefit contribution. Analysis of O₃ concentrations shows that temperature is the most significant factor affecting its concentration prediction, and as the O₃ concentration increases, T gradually changes from a negative to a positive benefit contribution. This study can provide a theoretical basis and new ideas for the study of the mechanism of statistical forecasting algorithms for the prediction of pollutant concentrations during high concentration pollution.

- (3) By optimising the simulation of PM_{2.5} and O₃ concentrations at various grid points in and around Beijing, a pollutant

concentration data set with a horizontal grid resolution of 6 km and a temporal resolution of 1 h was established. The optimized spatial distribution of pollutant concentrations has improved considerably compared to the WRF-Chem model simulation, and more consistent with the spatial distribution of observed concentrations of pollutants. In addition, the XGBoost algorithm optimises PM_{2.5} and O₃ concentrations best when compared with the SVR, LR, DTR and RF algorithms.

Author statement

Xingqin AN and Qingyong LI designed the experiment and guided the test process.

Jiangtao LI simulated, interpreted the data and wrote the paper.

Chao WANG analyzed the data and chart preparation.

Haomin YU, Xinyuan ZHOU and Yangli-ao GENG designed the XGBoost algorithm.

All the authors contributed to discussing results and commenting on the paper.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. The name and description of each features

Type	Number	Name	Description
Meteorological features	1	U	x-wind component
	2	V	y-wind component
	3	W	z-wind component
	4	T	perturbation potential temperature (theta-t0)
	5	P	perturbation pressure
	6	PB	base state pressure
	7	ALT	inverse density
	8	PSFC	surface pressure

(continued on next page)

(continued)

Type	Number	Name	Description
Pollutant features	9	T00	base state temperature
	10	Q2	specific humidity at 2 m
	11	PBLH	planetary boundary layer height
	12	CLDFRA	cloud fraction
	13	SWDOWN	downward short wave flux at ground surface
	14	GLW	downward long wave flux at ground surface
	15	HFX	upward heat flux at the surface
	16	PH_NO2	photolysis rates
	17	QVAPOR	water vapor mixing ratio
	18	QCLOUD	cloud water mixing ratio
	19	ACGRDFLX	accumulated ground heat flux
	20	HGT	terrain height
	21	REFL_10CM	radar reflectivity (lamda = 10 cm)
	22	SWNORM	normal short wave flux at ground surface
	23	SWDDIR	shortwave surface downward direct irradiance
	24	SWDDNI	shortwave surface downward direct normal irradiance
	25	SWDDIF	shortwave surface downward diffuse irradiance
	26	EMISS	surface emissivity
	27	QFX	upward moisture flux at the surface
	28	ACLHF	accumulated upward latent heat flux at the surface
	29	CO	CO concentration, units: ppm
	30	NO2	NO ₂ concentration, units: ppm
	31	O3	O ₃ concentration, units: ppm
	32	SO2	SO ₂ concentration, units: ppm
	33	ORGAJ	secondary organic concentration
	34	ORGBAJ	biogenic organic concentration
	35	SOA_RXT_G	SOA (secondary organic aerosol) from reaction of GLY
	36	SOA_RXT_M	SOA (secondary organic aerosol) from reaction of MGLY
	37	ORGPAJ	primary organic concentration
	38	SO4AJ	sulfate concentration
	39	NO3AJ	nitrate concentration
	40	NH4AJ	ammonium concentration
	41	ECJ	elemental carbon concentration
	42	CLAJ	aerosol chloride concentration
	43	P25AJ	unknown PM _{2.5} concentration
	44	NAAJ	aerosol sodium concentration
	45	DUST_1	dust aerosols with 0.73 μm effective radius
	46	DUST_2	dust aerosols with 1.40 μm effective radius
	47	DUST_3	dust aerosols with 2.40 μm effective radius
	48	DUST_4	dust aerosols with 4.50 μm effective radius
	49	DUST_5	dust aerosols with 8.00 μm effective radius
Time series features	50	CO_pp	CO concentration, units: μg/m ³
	51	NO2_pp	NO ₂ concentration, units: μg/m ³
	52	O3_pp	O ₃ concentration, units: μg/m ³
	53	SO2_pp	SO ₂ concentration, units: μg/m ³
	54	PM25_pp	PM _{2.5} concentration, units: μg/m ³
	55	PM10_pp	PM ₁₀ concentration, units: μg/m ³
Periodic features	56–79	u_tem_P_N	periodic mean characteristic, where P stands for PM _{2.5} , O ₃ , NO ₂ , CO, SO ₂ and PM ₁₀ ; N stands for 3, 6, 12 and 24
	80–103	sigma_tem_P_N	periodic variance characteristic, where P stands for PM _{2.5} , O ₃ , NO ₂ , CO, SO ₂ and PM ₁₀ ; N stands for 3, 6, 12 and 24
Social features	104–127	u_per_P_N	periodic mean characteristic, where P stands for PM _{2.5} , O ₃ , NO ₂ , CO, SO ₂ and PM ₁₀ ; N stands for 3, 7, 14 and 30
	128–151	sigma_per_P_N	periodic variance characteristic, where P stands for PM _{2.5} , O ₃ , NO ₂ , CO, SO ₂ and PM ₁₀ ; N stands for 3, 7, 14 and 30
	152	Working_hour	time period is from 8:00 to 18:00 Beijing time
	153	Working_day	from Monday to Friday
	154	season	divided into spring, summer, autumn and winter

Appendix B. The calculation formula of evs, mae, mse, meae and r

$$evs = 1 - \frac{Var\{y - \hat{y}\}}{Var\{y\}}$$

$$mae = \frac{1}{n} \sum_{i=1}^{n-1} |y_i - \hat{y}_i|$$

$$mse = \frac{1}{n} \sum_{i=1}^{n-1} (y_i - \hat{y}_i)^2$$

$$meae = \text{median}(|y_1 - \hat{y}_1|, |y_2 - \hat{y}_2|, \dots, |y_n - \hat{y}_n|)$$

$$r = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$$

where $\hat{y}_i$ is the predicted value of the i -th sample, and y_i is the corresponding observed result, Var is Variance. moreover, The evs computes the explained variance regression score, the mae computes mean absolute error, the mse computes mean square error, the mae is calculated by taking the median of all absolute differences between the target and the prediction, and the r computes the coefficient of determination.

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